

Hack The World(s)

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Agenda

- 1 What You Have & What We Expect
- 2 Presentation Requirements
- 3 Two Rooms Example
- 4 Timeline & Judging

Data & Project Tracks

Two kinds of data: the **datasets we provide** or **data you generate yourself**.
Pick a track below — or use one as a springboard for your own idea.

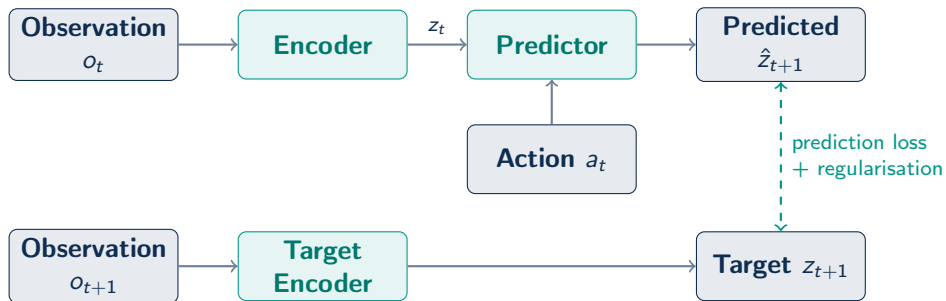
Group A — A new modality

- 1 Neuro - EEG
- 2 Audio / speech representation
- 3 Wearable biosignals (ECG / IMU / PPG)
- 4 Physical fields — *The Well*
- 5 Weather forecasting
- 6 Financial time series
- 7 Anomaly & predictive maintenance
- 8 3D point clouds / LiDAR

Group B — Vision & robotics

- 9 Learned cost / value for planning
- 10 Hierarchical / multi-timescale world model
- 11 Stress-test under factors of variation
- 12 Does intuitive physics emerge?

The Model — A JEPA Baseline



What we provide

A working **JEPA** that predicts *in representation space*, with regularisation losses to prevent collapse.
Three tracks: **image**, **video**, **action-conditioned video**.

Pipeline — one engine: swappable modules

One generic, config-driven JEPA engine in `eb_jepa/`; each example just wires swappable core modules.

Core engine — `eb_jepa/`

- `jepa.py` — JEPA/JEPAProbe, unified `unroll()`
- `architectures.py` — encoders / predictors (ResNet, Impala, GRU)
- `losses.py` — predictive JEPA & anti-collapse (VICReg / VC / IDM)
- `planning.py` — CEM / MPPI planners & objectives

Around it — data & assembly

- `datasets/<env>/` — loaders, dispatched by `env_name`
- `examples/<x>/` — thin scripts: `main.py` · `eval.py` · `config.yaml` · `README`
- `launch_sbatch.py` — one launcher (`-example`, `submitit`)
- `tests/` lock the swappable-module contracts

Pipeline — examples to reuse, tracks to complete, and above all: explore

Reuse a complete example, complete a track baseline by filling a tiny #TODO surface — then explore.

Reuse — complete examples

- `image_jepa` — CIFAR-10 SSL, linear probe ~91%
- `video_jepa` — Moving-MNIST latent prediction
- `ac_video_jepa/two_rooms` — world model + planning (~97%)
- *explore: **the maze** — A^* -free hierarchical navigation*

Complete — 6 tracks (baseline + #TODO)

- `audio` · `EEG` · `point cloud` · `FinTime` · `LTSF` · `Gray-Scott`
- **provided & runnable:** loader + augment + train loop + probe harness
- **you fill 3 #TODO stubs:** `build_encoder` · `build_ssl/build_jepa` · `probe`

“Start from a baseline, then make it yours”: runnable harness → fill the stubs → your baseline → explore.

What We Expect From You

Start from a baseline example, then make it yours.

01 A working model

Train a JEPA world model on the data of your choice.

- Keep it reproducible (`config.yaml`)
- Save at least one stable checkpoint

02 10-minute presentation

Walk us through your approach, experiments, and findings.

- Explain your architectural choices
- Show what the model learned

The Gold Standard

What makes a great submission?

- **Clear problem framing**
- **Quantitative results** vs. baseline
- **A genuine insight** or finding
- **Honest discussion** of limits
- **At least one ablation study**



What Your Presentation Should Cover

Your 10-minute presentation should address the points below.

Data → **Architecture** → **Training** → **Inference** → **Evaluation**

(plus optional **Bonuses** to go further)

Explain the data you trained on.

1 Where is it from?

- **Generated** by you
- **Chosen** from the provided datasets
- **Found elsewhere** (cite the source)

Modality, size, difficulty?

2 What does it look like?

Show it — stats, samples, a projection (PCA):



3 How did you prepare it?

- Normalisation
- Encoding / decoding target
- Data augmentation?

Any preprocessing worth noting?

Describe your model and how it was trained.

The model

- **Unchanged?** A quick JEPA recap
- **Modified?** Why, what, intuition, papers
- Number of parameters & modules

Loss & training dynamics

- Which (sub)losses? how balanced?
- Representation collapse — observed?
- Regularisation & stability



How did you train, and how did you iterate efficiently?

Setup

- Batch size, optimiser
- Learning-rate scheduler
- Number of epochs

Comparing without full training

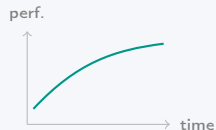
- A **proxy metric** to rank runs early
- A strategy to compare *without* a full run
e.g. scaling laws, ablations, ...

How do you use the trained model?

Inference strategy

- Method → Mode 1 / Mode 2, MCTS, actor, ...?
- Performance vs. inference time
- Any inference tricks?

Performance vs. time



more compute, better rollout?

How good is it, and how do you know?

Robustness

- Ablations
- Scaling laws
- Seed & training stability

Performance

- Dataset performance
- Compare to a baseline

Reach

- Application to real-life situations
- A step towards AGI?

Optional — ways to stand out.

Rewarded

- High performance
- Ablations Hyperparameter tuning
- Extensive training-stability analysis — esp. visualising representation collapse

Going further

- Handmade (non-synthetic) dataset
- Method that scales to other datasets of the field (e.g. time series)
- Hierarchical world models

Two Rooms Example¹

HACK THE WORLD - WORLD-MODELS TRACK

Two-Rooms World Model

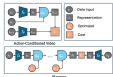
An action-conditioned JEPA for goal-reaching navigation

Team Organizer - June 2024 - Hackathon submission
Color code in this deck: **red** = a possible improvement **gold** = it would be highly useful

OVERVIEW - OUR SETUP

The example pipeline

Example submission - baseline EIP-JEPA, unchanged, trained on `Two_Rooms` via the `je_1x160k_1epa-ncjpa`



Baseline EIP-JEPA, action-conditioned training (top) & planning (bottom)

ARCHITECTURE & LOSS - 1/2

Model architecture

Unchanged baseline JEPA (ncjpa)

- Encoder: $\text{InputEmbedder} \rightarrow \text{latent} [B, L, T, H, W]$
- Action enc. MLP
- Predictor: RMSPredictor , autoregressive, old window +
- Regularizer: V_{diff} , KL_{diff} , KL_{diff}
- Predictor: SquashedGating

[JEPA paper](#) [ncjpa](#)

RECOMMENDATION Swapping $\text{RMSPredictor} \rightarrow$ a $\text{Transformer} / \text{diffusion predictor}$ (with papers + Δparams) could lift capacity.

RECOMMENDATION A hierarchical world model — multi-timescale predictor (slow over rooms, fast over steps) — would be highly valued.

RECOMMENDATION Generalizing the architecture to another modality (e.g. time-series) would really stand out.

BEFORE WE START

How to read this deck

It's a structure. A clear, section-by-section way to present the results of your study.

It's a source of ideas. The red & gold notes suggest ways to enrich your analysis and presentation.

It's not just to do

Don't try to implement everything suggested here. The core of your work should be your own research — searching for new regularizations, borrowing methods from other fields, trying fresh ideas. These notes are only about presenting good results excellently — moving a good submission to an outstanding one.

DATA - 1/2

Dataset selection

Generated (synthetic), `Two_Rooms` trajectories produced on-the-fly

- Methodology: procedural navigation layout + random action rollout
- Scalability: ~ 1000 GPU (1000) pipeline (double-buffered)
- Shape: $\text{obs} [B, L, T, H, W]$, $T \sim 17$, random policy
- Compute: 0 storage, generated per batch on-device

RECOMMENDATION Finding the most revealing visualization would help a lot — e.g. a PCA / t-SNE of the latents colored by true XY.

RECOMMENDATION Offline memmap would enable reproducible scaling-law sweeps over dataset size.



Two very sampled subsets in visual-observability heatmap

RECOMMENDATION Reporting difficulty levels (door width, asymmetry) and building an easy-hard split would strengthen the story.

ARCHITECTURE & LOSS - 2/2

Loss formulation & collapse

Total = $\text{ProdCost} + \lambda \text{Regularizer}$, where the regularizer contains:

- Variance + Covariance (anti-collapse)
- Temporal stability
- Inverse-dynamics (ID)

Baseline coeffs: cov_d , var_d , cov_t , var_t , cov_d , var_d (defaults)

Collapses considered on embedding dist, held-up by the variance term.

RECOMMENDATION A quantitative collapse metric (effective rank of the covariance, not just std) would make the claim solid.

RECOMMENDATION An extensive visual collapse analysis — covariance heatmaps across epochs — is highly valued.

RECOMMENDATION Sweeping each sub-loss coefficient against probe error would show the balance matters.

OVERVIEW - THE TASK

Goal-reaching in Two Rooms

- Add navigable two rooms joined by a door — the `TwoRooms` env.
- Legally: joint observations + actions.
- Learn a world model whose latent predicts future states.
- Then plan actions to reach a goal position.

RECOMMENDATION Framing the whole submission as a scaling study rather than a single run would be far more convincing than one number.



A single `TwoRooms` episode: start — goal across the door

DATA - 2/2

Preprocessing & target framing

Normalization: per-channel normalized via the `TwoRooms` coordinate

Encode target (Goal): ground truth frames — encoder — latent; the JEPA has lives in this latent space. Decode (Goal): train: JEPA uses MLP to map latent — XY position for measurable status. Augmentation: none in baseline

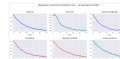


RECOMMENDATION Adding augmentation (action noise, random crops, color jitter) would be a cheap robustness win and an easy ablation.

RECOMMENDATION Justifying the target — why XY? — by showing the probe is linear-decodable would double as a collapse sanity check.

ARCHITECTURE & LOSS - 2/2

Training dynamics



Total = prediction cost and each regularizer sub-term tracked separately — all converging smoothly, no divergence

RECOMMENDATION Overlaying an unstable run (too-high LRT) beside the smooth one would nicely illustrate how hyperparameters drive stability.

⁰Found it at [hackathon_guide/two_rooms_example_slides.pdf](#)

Schedule

17:30 Code submission

18:00 Slides submission & pre-jury

19:00 Final jury (*qualified teams*)



Judging Criteria

- **Technical quality**

Robustness, code cleanliness & architecture

- **Results**

Performance, metrics & meaningful comparisons

- **Presentation**

Clarity, storytelling & visual aids

- **Originality**

Ablations, novel insights & going the extra mile

Now go build your world.

Good luck, and have fun!